

Measuring Unruh temperatures

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Abstract

Conventional wisdom holds that no practical experimental setup will be able to measure the Unruh temperature of any realistic physical system in the foreseeable future. This conclusion is inescapable if the Unruh temperature is understood in the narrow sense of a QFT effect experienced by eternally accelerating observers. In this paper we propose that adopting a broader point of view where quantum entanglement is the fundamental physical phenomenon underlying the appearance of a temperature may lead to an actual measurement of this temperature in the laboratory. Our point of view is not new – a relationship between entanglement and entropy in particular has been suggested by several groups – but the prediction of a measurable temperature in the tens of μK appears to be new.

OVERVIEW OF UNRUH EFFECT

The Unruh effect [1–3] involves an observer moving with constant proper acceleration A (a Rindler observer). In a 2-d flat spacetime, this observer may be characterized by the relationships ($c = 1$)

$$x = \alpha e^{X/\alpha} \cosh \frac{T}{\alpha} \quad ; \quad t = \alpha e^{X/\alpha} \sinh \frac{T}{\alpha} \quad (1)$$

where $\alpha > 0$ is a parameter with dimension of length, (x, t) are the coordinates used by the inertial observer, and (X, T) the corresponding coordinates used by the accelerating observer. The Rindler observer at a fixed position X_0 has constant proper acceleration $A = \alpha^{-1} e^{-X_0/\alpha}$, and the (X, T) coordinates cover only the right Rindler wedge in spacetime [4].

The quantization of a massless scalar field Φ is conveniently carried out using light cone coordinates $u = t - x$, $v = t + x$, $U = T - X$, and $V = T + X$. Because left-moving and right-moving modes do not interact, we may study them independently. In particular, we have for the left-moving modes of our scalar field operator

$$\int_0^\infty \frac{d\omega}{\sqrt{4\pi\omega}} (\hat{a}_\omega e^{-i\omega v} + \hat{a}_\omega^\dagger e^{i\omega v}) = \int_0^\infty \frac{d\Omega}{\sqrt{4\pi\Omega}} (\hat{b}_\Omega e^{-i\Omega V} + \hat{b}_\Omega^\dagger e^{i\Omega V}) \quad (2)$$

Although limited to the right Rindler wedge, this relationship suffices to solve for the operators $\hat{b}_\Omega$ and $\hat{b}_\Omega^\dagger$ used by the accelerated observer. The result takes the form of a

Bogoliubov transformation,

$$\hat{b}_\Omega = \int_0^\infty d\omega (\alpha_{\omega\Omega} \hat{a}_\omega + \beta_{\omega\Omega} \hat{a}_\omega^\dagger) \quad (3)$$

with

$$\alpha_{\omega\Omega} = \frac{1}{2\pi} \sqrt{\frac{\Omega}{\omega}} \int_{-\infty}^\infty dV e^{-i\omega v + i\Omega V} \quad (4)$$

and

$$\beta_{\omega\Omega} = \frac{1}{2\pi} \sqrt{\frac{\Omega}{\omega}} \int_{-\infty}^\infty dV e^{i\omega v + i\Omega V} \quad (5)$$

As may be seen by going to the complex plane (for details, see Refs. [4] or [5]), the Bogoliubov coefficients are related in a simple manner:

$$\alpha_{\omega\Omega} = e^{\pi\Omega\alpha} \beta_{\omega\Omega} \quad (6)$$

The expectation value of the number operator $N_\Omega = \hat{b}_\Omega^\dagger \hat{b}_\Omega$ in the Minkowski vacuum $|0\rangle_M$ can now be determined from the above results and the commutators of the basic operators. With $\langle N_\Omega \rangle = {}_M\langle 0 | N_\Omega | 0 \rangle_M$, and keeping in mind that $\hat{a}_\omega |0\rangle_M = 0$, we have

$$\langle N_\Omega \rangle = {}_M\langle 0 | \int_0^\infty d\omega \beta_{\omega\Omega}^* \hat{a}_\omega \int_0^\infty d\omega' \beta_{\omega'\Omega} \hat{a}_{\omega'}^\dagger | 0 \rangle_M \quad (7)$$

Using the commutator $[\hat{a}_\omega, \hat{a}_{\omega'}^\dagger] = \delta(\omega - \omega')$, this becomes

$$\langle N_\Omega \rangle = \int_0^\infty d\omega |\beta_{\omega\Omega}|^2 \quad (8)$$

As an alternative, we may use the commutator $[\hat{b}_\Omega, \hat{b}_{\Omega'}^\dagger] = \delta(\Omega - \Omega')$ to write $N_\Omega = \hat{b}_\Omega \hat{b}_\Omega^\dagger - \delta(0)$. The presence of the $\delta(0)$ term reflects the use of an infinite volume and is easily avoided in a finite volume formulation where only discretized frequencies are possible. Following the same steps as above, we get

$$\langle N_\Omega \rangle = \int_0^\infty d\omega |\alpha_{\omega\Omega}|^2 - \delta(0) \quad (9)$$

Combining (6), (7), and (8) leads to a Bose-Einstein-like distribution for $\langle N_\Omega \rangle$,

$$\langle N_\Omega \rangle = \frac{\delta(0)}{e^{2\pi\Omega\alpha} - 1} \quad (10)$$

or, for the average number density $\langle n_\Omega \rangle$

$$\langle n_\Omega \rangle = \frac{1}{e^{2\pi\Omega\alpha} - 1} \quad (11)$$

This familiar result is often interpreted as evidence of a thermal distribution of particles at the temperature $T = \hbar/2\pi k\alpha$. But this is clearly misleading unless the state at hand is thermal. Our next task becomes obvious if we recall that the analysis so far makes use of the right Rindler wedge only: without an extension of the accelerated coordinate system to all spacetime it is impossible to prove that the state $|0\rangle_M$ appears as a thermal state to the Rindler observer. And absent that proof, there is no basis for the claim that $T = \hbar/2\pi k\alpha$ is a physical temperature.

The argument leading from the extension of coordinates to the relationship between the Minkowski vacuum $|0\rangle_M$ and the Rindler vacuum $|0\rangle_R$ is well known (see, for instance, Refs. [4] and [5]) and will not be repeated here. For the purposes of this paper, it is sufficient to concentrate on the steps that highlight the physical underpinnings of the Unruh effect.

The introduction of a second coordinate patch to complement (1) brings with it a second set of operators $\hat{b}_\Omega^{(L)}$, $\hat{b}_\Omega^{(L)\dagger}$. We relabel the original set of operators above $\hat{b}_\Omega^{(R)}$, $\hat{b}_\Omega^{(R)\dagger}$ for clarity, and define the Rindler vacuum $|0\rangle_R$ by $\hat{b}_\Omega^{(L)}|0\rangle_R = \hat{b}_\Omega^{(R)}|0\rangle_R = 0$. In close parallel to the discussion above, we find that the operators $\hat{b}_\Omega^{(L)}$ and $\hat{b}_\Omega^{(R)}$ are given by Bogoliubov transformations as in (3), with $\hat{b}_\Omega^{(L)}$ involving the complex conjugates $\alpha_{\omega\Omega}^*$ and $\beta_{\omega\Omega}^*$. The Bogoliubov coefficients themselves continue to obey (6). With these results at hand it is not difficult to prove that $N_\Omega^{(L)}|0\rangle_M = N_\Omega^{(R)}|0\rangle_M$, that is, an expansion of $|0\rangle_M$ in eigenkets of $N_\Omega^{(L)}$ and $N_\Omega^{(R)}$ can only involve equal numbers of L and R particles. Indeed, as shown in Refs. [4, 5],

$$|0\rangle_M = Z^{-1/2} \prod_i \sum_{n_i=0}^{\infty} e^{-n_i \pi \Omega_i \alpha} |n_i\rangle_L |n_i\rangle_R \quad (12)$$

with Z the normalization factor

$$Z^{-1} = {}_R\langle 0|0\rangle_M = \prod_i (1 - e^{-2\pi \Omega_i \alpha}) \quad (13)$$

where we have switched from continuous to discrete frequencies for simplicity.

The squeezed-state nature of the Minkowski vacuum becomes apparent if we define ($\hat{b}_i \equiv \hat{b}_{\Omega_i}$)

$$S = \prod_i S(\xi_i) = \prod_i e^{\xi_i (\hat{b}_i^{(L)\dagger} \hat{b}_i^{(R)\dagger} - \hat{b}_i^{(L)} \hat{b}_i^{(R)})} \quad (14)$$

Then, letting

$$\tanh \xi_i = e^{-\pi \Omega_i \alpha} \quad (15)$$

we have [5]

$$|0\rangle_M = S|0\rangle_R \quad (16)$$

Since left wedge degrees of freedom are inaccessible to the Rindler observer, measurements carried out by this observer require a tracing out over such degrees of freedom. The Rindler observer therefore uses the reduced density matrix

$$\rho_R = \text{Tr}_L \rho = \text{Tr}_L |0\rangle_{MM} \langle 0| \quad (17)$$

to compute expectation values of observables,

$$\langle A \rangle = \text{Tr}_R \rho_R A \quad (18)$$

Although ρ is the density matrix for a pure state, ρ_R represents a thermal state, as is clear from [5]

$$\rho_R = Z^{-1} \prod_i \sum_{n_i=0}^{\infty} e^{-2n_i \pi \Omega_i \alpha} |n_i\rangle_{RR} \langle n_i| \quad (19)$$

The non-pure nature of ρ_R is also evident from

$$\text{Tr}_R \rho_R^2 = Z^{-2} \prod_i \sum_{n_i=0}^{\infty} e^{-4n_i \pi \Omega_i \alpha} = \prod_i \tanh \pi \Omega_i \alpha < 1 \quad (20)$$

The thermal nature of ρ_R justifies the identification of $T = \hbar/2\pi k\alpha$ with the temperature of the state as measured by the Rindler observer. Furthermore, the steps outlined above single out the erasure of $L - R$ entanglement information in (17) as the physical origin of the temperature [6].

MECHANICAL OSCILLATORS AND UNRUH TEMPERATURE

In this section we propose to take advantage of recent progress in the construction of two-mode squeezed states in mechanical oscillators to validate the conceptual framework outlined above and to measure the equivalent Unruh temperature of the system. More specifically, we focus on the work of Ref. [7] with atoms in a two-dimensional optical lattice.

We emphasize at the outset that this section involves nonrelativistic physics only. The role of special relativity in the QFT Unruh effect is limited to providing the horizons that force the tracing out of left-wedge degrees of freedom. This is replaced here with an experimental

choice involving the degrees of freedom one wishes to measure in the two-dimensional optical lattice.

Consider a non-interacting two-dimensional isotropic oscillator with Hamiltonian

$$H_A = H_x + H_y \quad (21)$$

with $(i = x, y)$

$$H_i = \hbar\omega \left(a_i^\dagger a_i + \frac{1}{2} \right) \quad (22)$$

and denote the ground state of H_A by $|0\rangle_A \equiv |n_x = 0\rangle \otimes |n_y = 0\rangle$. In analogy with (14), let

$$S(\xi) = \exp(\xi a_x^\dagger a_y^\dagger - \xi^* a_x a_y) \quad (23)$$

where $\xi = r e^{i\phi}$ is allowed to be a complex number for generality. The squeezing operator (23) relates the annihilation operators a_x and a_y to a new set of operators b_1 and b_2

$$a_x = S b_1 S^\dagger = \cosh r b_1 - e^{i\phi} \sinh r b_2^\dagger \quad (24)$$

$$a_y = S b_2 S^\dagger = \cosh r b_2 - e^{i\phi} \sinh r b_1^\dagger \quad (25)$$

The corresponding Hamiltonian is

$$H_B = \hbar\omega \left(b_1^\dagger b_1 + b_2^\dagger b_2 + 1 \right) \quad (26)$$

and the ground states are related by

$$|0\rangle_B = S^\dagger |0\rangle_A \quad (27)$$

We shall view $|0\rangle_A$ as the analog of $|0\rangle_R$, and $|0\rangle_B$ as the analog of $|0\rangle_M$.

The definitions (24) and (25) imply

$$a_x |0\rangle_B = -e^{i\phi} \tanh r a_y^\dagger |0\rangle_B \quad (28)$$

$$a_y |0\rangle_B = -e^{i\phi} \tanh r a_x^\dagger |0\rangle_B \quad (29)$$

from where it follows that

$$N_x |0\rangle_B = N_y |0\rangle_B \quad (30)$$

as in the Unruh case. This is an indication of correlations between the x and y modes in the $|0\rangle_B$ ground state, which may corroborated by computing the expectation value

$${}_B\langle 0 | a_x a_y | 0 \rangle_B = -\frac{1}{2} e^{i\phi} \sinh 2r.$$

It is also easy to see from (24) and (25) that

$${}_B\langle 0| N_x |0\rangle_B = {}_B\langle 0| N_y |0\rangle_B = \sinh^2 r \quad (31)$$

This is a well-known result in quantum optics. For our present purposes, it is more convenient to introduce an alternative squeezing parameter ζ by

$$\tanh^2 r = e^{-\zeta} \quad (32)$$

In terms of ζ , (31) becomes

$${}_B\langle 0| N_x |0\rangle_B = {}_B\langle 0| N_y |0\rangle_B = \frac{1}{e^\zeta - 1} \quad (33)$$

in close correspondence with the Bose-Einstein distribution (11) found in the Unruh case. As in that case, we cannot yet interpret this as evidence that our system is thermal, since the density matrix $\rho = |0\rangle_B {}_B\langle 0|$ is obviously pure.

If only x modes are accessible, the appropriate density matrix will be

$$\rho_x = \text{Tr}_y \rho = \text{Tr}_y |0\rangle_B {}_B\langle 0| \quad (34)$$

and for the simple case at hand, the expansion

$$|0\rangle_B = \sum_{n=0}^{\infty} c_n |n, n\rangle_A \quad (35)$$

with

$$c_n = \frac{(-e^{i\phi} \tanh r)^n}{\cosh r} \quad (36)$$

and $|n, n\rangle_A = |n_x = n\rangle_A \otimes |n_y = n\rangle_A$ is well known (see, e.g., [8, 9]). It follows that the reduced density matrix (34) is

$$\rho_x = \frac{1}{\cosh^2 r} \sum_{n=0}^{\infty} (\tanh r)^{2n} |n\rangle_{AA} \langle n| = \frac{1}{\cosh^2 r} \sum_{n=0}^{\infty} e^{-n\zeta} |n\rangle_{AA} \langle n| \quad (37)$$

The thermal nature of ρ_x becomes obvious if we define

$$\tilde{H}_x = \hbar \omega a_x^\dagger a_x \quad (38)$$

and write

$$\rho_x = \frac{1}{Z} \sum_{n=0}^{\infty} e^{-\beta \tilde{H}_x} |n\rangle_{AA} \langle n| \quad (39)$$

where $Z = \text{Tr } e^{-\beta \tilde{H}_x} = \cosh^2 r$ is the partition function, and $\beta = \zeta/\hbar\omega$. The latter leads to an effective temperature for this system,

$$T = \frac{\hbar\omega}{2k \ln \coth r} \quad (40)$$

For an estimate of this temperature, we may approximate the experimental parameters in [7] as $r \approx 1$ and $\omega \approx 1\text{MHz}$ to obtain $T \approx 1.4 \times 10^{-5}$ K. Although this is a low temperature, it is by no means out of reach, and should be contrasted with an Unruh temperature of $T \sim 10^{-19}$ K for an acceleration $a \sim 10 \text{ m/s}^2$. A measurement of the temperature (40) would not, of course, provide empirical verification of the Unruh effect interpreted in the narrow sense of a QFT phenomenon linking temperature to acceleration. But it would certainly provide support for the more general notion that quantum entanglement (or rather, its erasure) is the essential physical mechanism giving rise to a temperature in certain systems [6, 9].

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